

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12414895
B2
September 16, 2025
St-Pierre; Louis et al.

Weight support device and apparatus

Abstract

A weight support device for use with a walking aid to at least partially support a weight of a user. The weight support device includes an elongated guiding body mountable to the walking aid and having a longitudinal axis extending vertically, a connection body slidably mounted to the elongated guiding body for translating along the longitudinal axis, at least one constant force spring rotatably mounted to the elongated guiding body with an outer end of the spring connected to the connection body, and a body securing device for engagement with the user and mounted to the connection body. When the elongated guiding body is mounted to the walking aid, the spring exerts a restoring constant force on the connection body when the connection body moves away from the first end of the elongated guiding body.

Inventors: St-Pierre; Louis (Montreal, CA), Kamdem; Yoann (Sainte-Cecile-les-Vignes, FR), Jones; Guillaume (Montreal, CA), Lambert; Sarah (Montreal, CA), Michaud; Marie-Pier (Montreal, CA), Lambert; Samuel (Sainte-Barbe, CA)

Applicant: ORA MÉDICAL INC. (Montreal, CA)

Family ID: 1000008818687

Assignee: ORA MÉDICAL INC. (Montreal, CA)

Appl. No.: 18/569646

Filed (or PCT Filed): June 17, 2022

PCT No.: PCT/IB2022/055645

PCT Pub. No.: WO2022/264103

PCT Pub. Date: December 22, 2022

Prior Publication Data

Related U.S. Application Data

us-provisional-application US 63202591 20210617

Publication Classification

Int. Cl.: A61H3/00 (20060101); A61H3/04 (20060101)

U.S. Cl.:

CPC A61H3/008 (20130101); A61H3/04 (20130101); A61H2003/007 (20130101); A61H2201/0192 (20130101); A61H2201/163 (20130101)

Field of Classification Search

CPC: A61H (3/008); A61H (2003/007); A63B (22/02-0292)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8151812	12/2011	Razon	297/195.1	A61H 3/04
9914003	12/2017	Kuehne	N/A	A63B 71/0009
2001/0048206	12/2000	Niu et al.	N/A	N/A
2010/0170546	12/2009	Popovic	135/67	A61H 3/008

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3021867	12/2018	FR	N/A

OTHER PUBLICATIONS

International Search Report; Canadian Intellectual Property Office; International Application No. PCT/IB2022/055645; Sep. 21, 2022; 3 pages. cited by applicant

Written Opinion of the International Searching Authority; Canadian Intellectual Property Office; International Application No. PCT/IB2022/055645; Sep. 21, 2022; 3 pages. cited by applicant

Primary Examiner: Robertson; Jennifer

Attorney, Agent or Firm: Taft Stettinius & Hollister LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a U.S. National Phase of International PCT Application No. PCT/IB2022/055645 filed Jun. 17, 2022, which claims

FIELD

(1) The present technology pertains to the field of medical devices for rehabilitation of subjects suffering from locomotor disabilities, and more particularly to weight support devices for walking aids.

BACKGROUND

(2) Rehabilitation allows subjects suffering from locomotor issues to improve their mobility. Usually, the rehabilitation is performed through physical exercises under the supervision of a healthcare professional. However, not all subjects have access to such a healthcare professional or the access they have is limited. To compensate for the limited access to a healthcare professional, walking aids provided with a weight support device have been developed to allow a subject to perform rehabilitation exercises without the help of a healthcare professional. However, most of the weight support devices present at least one of the following disadvantages. do not adequately support the weight of the subject. At least some of the weight support devices do not offer adequate support for rehabilitation, have a design that forces a subject to perform gestures inadequate for rehabilitation, and/or are expensive.

(3) Therefore, there is a need for an improved weight support device.

SUMMARY

(4) According to a first broad aspect, there is provided a weight support device for use with a walking aid to at least partially support a weight of a user, the weight support device comprising: an elongated guiding body extending between a first and a second end along a longitudinal axis and laterally between a front face and a rear face, the elongated guiding body being mountable to the walking aid, the longitudinal axis extending substantially vertically when the elongated guiding body is mounted to the walking aid and the walking aid is positioned on a horizontal surface and the first end being higher than the second end when the elongated guiding body is mounted to the walking aid; a connection body slidably mounted to the elongated guiding body for translating along the longitudinal axis of the elongated guiding body; at least one constant force spring rotatably mounted to the elongated guiding body, an outer end of the at least one constant force spring being connected to the connection body; and a body securing device for engagement with the user, the body securing device being mounted to the connection body, wherein when the elongated guiding body is mounted to the walking aid, the at least one constant force spring exerts a restoring constant force on the connection body when the connection body moves away from the first end of the elongated guiding body so as to compensate at least partially for the weight of the user.

(5) In one embodiment, the connection body is mounted on the front face of the elongated guiding body.

(6) In one embodiment, the at least one constant force spring is mounted at the first end of the elongated guiding body.

(7) In one embodiment, the outer end of the at least one constant force spring is secured to the connection body.

(8) In one embodiment, the outer end of the at least one constant force spring is removably secured to the connection body.

(9) In one embodiment, the elongated guiding body comprises a rail extending along the longitudinal axis and the connection body comprises a trolley slidably mounted to the rail.

(10) In one embodiment, the connection body further comprises a connection plate mounted to the trolley, the outer end of the at least one constant force spring being connected to the connection plate.

(11) In one embodiment, the connection body comprises a joint assembly securable to the body

securing device.

(12) In one embodiment, the at least one constant force spring is mounted on the rear face of the elongated guiding body, the weight support device further comprising a connection assembly extending between a first assembly end connected to the connection body and a second assembly end connected to the outer end of the at least one constant force spring, the connection assembly comprising at least a flexible section to engage the first end of the elongated guiding body.

(13) In one embodiment, the elongated guiding body comprises a front rail extending along the longitudinal axis and the connection body comprises a front trolley slidably mounted to the front rail.

(14) In one embodiment, the connection assembly comprises an elongated flexible body connected between the outer end of the at least one constant force spring and the connection body.

(15) In one embodiment, the connection assembly comprises a first connector and a second connector removably securable together, the first connector being connected to the connection body and the second connector being connected to the outer end of the at least one constant force spring.

(16) In one embodiment, the elongated guiding body comprises a rear rail extending along the longitudinal axis and the connection assembly further comprises a rear trolley slidably mounted to the rear rail, the rear trolley being connected to the connection body and the first connector.

(17) In one embodiment, the front trolley comprises a connection plate.

(18) In one embodiment, the connection body comprises a first rotational connection mounted to the connection plate and a second rotational connection mounted to the first rotational connection, the body securing device being mounted to the second rotational connection.

(19) In one embodiment, the at least one constant force spring is mounted adjacent the second end of the elongated guiding body.

(20) In one embodiment, the body securing device comprises a harness.

(21) According to another broad aspect there is provided a walking aid system comprising: a walking aid; and the above-described weight support device, the weight support device being secured to the walking aid so that the longitudinal axis extends substantially vertically when the walking aid is positioned on the horizontal surface.

(22) In one embodiment, the walking aid comprises one of a walker and a treadmill.

(23) In one embodiment, the weight support device offers new perspectives for the rehabilitation of subjects having locomotor disabilities. The weight support offered by a constant force spring provides a more adapted support in comparison to a usual spring and the contact force spring may be adapted based on the progress of the subject in the rehabilitation. The present weight support device allows for replicating the support provided by a healthcare professional to continue the rehabilitation outside a rehabilitation center or clinic. The present weight support device also allows for a healthcare professional to follow up the rehabilitation of a subject and the follow-up can be performed remotely.

(24) In one embodiment, the weight support device is integrated into a walking aid such as a wheeled walker, powered walker or the like. In one embodiment, a walking aid consists of an assistive device that intends to improve the walking pattern of a user. It reduces the weight bearing on affected or injured limbs and transfers the weight to the device, thereby reducing the pain suffered by the user and improving the user's ability to walk. The use of a walking aid also improves the balance of the user while walking by increasing the base of support. An example of a walking aid consists in a walker which usually comprises a frame and four points of contact with the ground.

(25) Implementations of the present technology each have at least one of the above-mentioned objects and/or aspects, but do not necessarily have all of them. It should be understood that some aspects of the present technology that have resulted from attempting to attain the above-mentioned object may not satisfy this object and/or may satisfy other objects not specifically recited herein.

(26) Additional and/or alternative features, aspects and advantages of implementations of the

present technology will become apparent from the following description, the accompanying drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:
- (2) FIG. 1 is a perspective view of a walker provided with a weight support device and a harness, in accordance with a first embodiment;
- (3) FIG. 2 is a perspective view of the weight support device and the harness of FIG. 1, the weight support device comprising an elongated guiding structure, a connection structure and a spring assembly;
- (4) FIG. 3 is a perspective view of the elongated guiding structure and part of the connection structure of the weight support device of FIG. 1;
- (5) FIG. 4 is a perspective exploded view of the part of the elongated guiding structure and part of the connection structure of the weight support device of FIG. 1;
- (6) FIG. 5 is a perspective exploded view of the spring assembly of the weight support device of FIG. 1;
- (7) FIG. 6 is a perspective view of the spring assembly, part of the elongated guiding structure and the connection structure of the weight support device of FIG. 1;
- (8) FIGS. 7a and 7b illustrate two springs of the spring assembly contained in the weight support assembly of FIG. 1 in a retracted position and an extended position, respectively, in accordance with an embodiment;
- (9) FIG. 8 is a perspective view of part of the connection assembly of the weight support device of FIG. 1 connected to the harness of FIG. 1;
- (10) FIG. 9 is a perspective view of a ball joint assembly contained in the connection structure of the weight support device of FIG. 1, in accordance with an embodiment;
- (11) FIG. 10 is a cross-sectional view of an arm contained in the ball joint assembly of FIG. 9;
- (12) FIG. 11 is a front perspective view of a walker provided with a weight support device and a harness, in accordance with a second embodiment;
- (13) FIG. 12 is a rear perspective view of the walker of FIG. 11;
- (14) FIG. 13 is a rear view of the walker of FIG. 11;
- (15) FIG. 14 is a left view of the walker of FIG. 11;
- (16) FIG. 15 is a front perspective view of the weight support device of the walker of FIG. 11 in a rest position;
- (17) FIG. 16 is a rear perspective view of the weight support device of FIG. 15;
- (18) FIG. 17 is a front view of the weight support device of FIG. 15;
- (19) FIG. 18 is a rear view of the weight support device of FIG. 15;
- (20) FIG. 19 is a perspective exploded view of the weight support device of FIG. 15;
- (21) FIG. 20 is a perspective exploded view of a spring assembly contained in the weight support device of FIG. 15, in accordance with an embodiment;
- (22) FIG. 21 is a front perspective view of the weight support device of FIG. 15 in an exemplary extended position;
- (23) FIG. 22 is a rear perspective view of the weight support device of FIG. 21;
- (24) FIG. 23 is a front view of the weight support device of FIG. 21;
- (25) FIG. 24 is a rear view of the weight support device of FIG. 21;
- (26) FIG. 25 is a front perspective view of the harness contained in the walker of FIG. 11, in

accordance with an embodiment; and

(27) FIG. 26 is an exploded perspective view of the harness of FIG. 25.

DETAILED DESCRIPTION

(28) The examples and conditional language recited herein are principally intended to aid the reader in understanding the principles of the present technology and not to limit its scope to such specifically recited examples and conditions. It will be appreciated that those skilled in the art may devise various arrangements which, although not explicitly described or shown herein, nonetheless embody the principles of the present technology and are included within its spirit and scope.

(29) Furthermore, as an aid to understanding, the following description may describe relatively simplified implementations of the present technology. As persons skilled in the art would understand, various implementations of the present technology may be of a greater complexity.

(30) In some cases, what are believed to be helpful examples of modifications to the present technology may also be set forth. This is done merely as an aid to understanding, and, again, not to define the scope or set forth the bounds of the present technology. These modifications are not an exhaustive list, and a person skilled in the art may make other modifications while nonetheless remaining within the scope of the present technology. Further, where no examples of modifications have been set forth, it should not be interpreted that no modifications are possible and/or that what is described is the sole manner of implementing that element of the present technology.

(31) FIG. 1 illustrates one embodiment, a walker 10 provided with a weight support device 12 and a harness 14. It should be understood that the illustrated walker 10 is exemplary only and the weight support device 12 can be used in connection with any other adequate walker, any adequate walking aid, any adequate treadmill or the like.

(32) The exemplary walker 10 comprises a U-shaped bar or rod 20, a transverse bar 22, two curved bars 24 and 26, four wheels 30, 32, 34 and 36 and two reinforcement arms 38 and 40. The wheels 30 and 32 are rotatably secured to a respective end 42, 44 of the U-shaped bar 20 while the wheels 34 and 36 are rotatably secured to a first end 46, 48 of a respective curved bar 24, 26.

(33) As illustrated in FIG. 1, the U-shaped bar 20 comprises two arm portions 50 and 52 each rotatably secured to a respective wheel 30, 32 and spaced apart by a central portion 54. A second end 56 of the curved bar 24 is secured to the arm portion 50 of the U-shaped bar 20 and a second end 58 of the curved bar 26 is secured to the arm portion 52 of the U-shaped bar 20. The transverse bar 22 is secured to the two arm portions 50 and 52 of the U-shaped bar 20 and extends therebetween. The reinforcement bar 38 extends between a first end secured to the arm portion 50 of the U-shaped bar 20 and a second end secured to the curved arm 24. The reinforcement bar 40 extends between a first end secured to the arm portion 52 of the U-shaped bar 20 and a second end secured to the curved arm 26.

(34) In the following, the space 56 between the curved bars 24 and 26 and the U-shaped bar 20 in which the harness 14 is positioned is referred to as the receiving space, i.e., the space for receiving a user of the walker 10.

(35) The weight support device 12 is secured to the walker 10, and more precisely to the transverse bar 22 and the central portion 54 of the U-shaped bar 20 of the walker 10, as described in greater detail below.

(36) The weight support device 12 comprises an elongated guiding structure or body 60 secured to the walker 10, a connection structure or body 62 and two spring assemblies 64. The elongated guiding structure 60 is fixedly secured to the transverse bar 22 and the central portion 54 of the U-shaped bar 20 and extends therebetween. The connection structure 62 is secured to the harness 14 and slidably connected to the elongated guiding structure 60. Each spring assembly 64 is fixedly secured to the top portion of the elongated guiding structure 60 and comprises two constant force springs that are operatively connected to the connection body 62 to each exert a constant restoring force on the connection structure 62 while they are extended due to the motion of the connection structure 62 away from the spring assembly 64.

(37) As illustrated in FIGS. 2 and 3, the elongated guiding structure **60** comprises a main elongated body **70** having the shape of a rod or tube, and a rail **72**. The main elongated body **70** extends longitudinally between a first or top end and a second or bottom end. The bottom end of the main elongated body **70** is fixedly secured to the transverse bar **22** of the walker **10** and the top portion of the main elongated body **70** is secured to the central portion **54** of the U-shaped bar **20**. The main elongated body **70** comprises a front face that faces the receiving space **56** and a back face opposite the front face. The rail **72** is fixedly secured to the front face and extends along at least a section thereof.

(38) As illustrated in FIGS. 2-4, the connection structure **62** comprises a trolley **80**, a connection plate **82**, two connectors **84** and a ball joint assembly **88**. The trolley **80** is operatively connected to the rail **72** so as to allow a translation of the trolley **80** along the rail **72**. The person skilled in the art will understand that the design of the trolley **80** and the design of the rail **72** may vary as long as a single degree of freedom exists between the rail **72** and the trolley **80**, i.e. as long as the trolley **80** can only translate along the rail **72**. In the illustrated embodiment, the rail **72** is provided with a recess extending along each one of its lateral faces and the trolley **80** is provided with a U-shaped cross-section. Protrusions that match the recesses of the rail **72** are provided on opposite internal faces of the trolley **80** and when the trolley **80** is connected to the rail **72**, each protrusion of the trolley **80** is inserted into a respective recess of the rail, thereby allowing only a translation of the trolley along the rail **72**.

(39) The connection plate **82** is secured to the trolley **80**. It will be understood that any adequate securing means for securing the connection plate **82** to the trolley **80** may be used. In one embodiment, the connection plate **82** is removably secured to the trolley **80**. For example, screws may be used for securing the connection plate **82** to the trolley **80**. In another embodiment, the connection plate is fixedly secured to the trolley **80**.

(40) The connection plate **82** is provided with two holes or apertures **86** which are located adjacent to a top end of the connection plate **82**. As described in greater detail below, the holes **86** are used for securing the constant force springs to the connection plate **82**.

(41) In one embodiment, the weight support device **12** further comprises a brake trolley **87** of which the position along the rail **72** can be fixed. Similarly to trolley **80**, trolley **87** is operatively connected to the rail **72** so as to only allow a translation of the trolley **87** along the rail **72**. In the illustrated embodiment, trolley **87** is provided with a U-shaped cross-section. Protrusions that match the recesses of the rail **72** are provided on opposite internal faces of the trolley **87** and when the trolley **87** is connected to the rail **72**, each protrusion of the trolley **80** is inserted into a respective recess of the rail, thereby allowing only a translation of the trolley **87** along the rail **72**. The trolley **87** further comprises a brake mechanism that when activated prevents the trolley **87** from translating along the rail **72**. In order to limit the translation of the trolley **80** along the rail **72**, the trolley **87** can be positioned at a desired position along the rail **72** and the brake mechanism is actuated so as to fix the trolley at the desired position by preventing any translation of the trolley **87** along the rail **72**. The fixed position of the brake trolley along the rail **72** then limits the possible translation of the trolley **80** along the rail **72** by defining an extreme possible position for the trolley **80**.

(42) FIG. 5 illustrates one embodiment of a spring assembly **64**. A spring assembly **64** comprises a base plate **90**, a casing **92**, a cover plate **94** and two constant force springs **96**. It will be understood that in the illustrated embodiment, the two spring assemblies **64** illustrated in FIG. 2 share the same base plate **90**. Additionally, only one spring assembly **64** can be mounted on the single base plate **90** if a weaker restoring force is required for a specific patient.

(43) The base plate **90** is securable to the top end of the elongated body **70**. In the illustrated embodiment, the elongated body **70** is hollow and the base plate **90** comprises a protrusion **98** that projects from a bottom face thereof. The protrusion of the base plate **90** snugly fits into the hollow elongated body **70** so as to secure the base plate to the elongated body **70**. It will be

understood that any adequate means for securing the base plate **90** to the elongated body **70** may be used. For example, adhesive, screws, etc. may be used for securing the base plate **90** to the elongated body **70**.

(44) The base plate **90** is further provided with two recesses **100** in which at least the outer end of the constant force spring **96** may extend.

(45) The constant force spring **96** is mounted on a shaft **102** for rotatably securing the constant force spring **96** to the casing **92** and the cover plate **94**. A tubular spacer **104** is mounted about the shaft **102** and a bearing **106** is mounted about the spacer **104**. The constant force spring **96** is mounted about the bearing **106**. The casing **92** is provided with a first aperture **110** on a lateral face thereof and the cover plate **94** is provided with a second aperture **112**. The apertures **110** and **112** are designed so as to each receive a respective end of the shaft **102** so as to rotatably secure the constant force spring **96** to the casing **92** and the cover plate **94**, and therefore to the elongated guiding structure **60**.

(46) The constant force spring **94** comprises a wound or coil section connected to a flat outer end. A constant force spring usually consists of a pre-stressed flat strip of spring material which is formed into substantially constant radius coils around itself or on a drum. When the strip is extended (deflected) the inherent stress resists the loading force, the same as a common extension spring, but at a nearly constant (zero) rate. A substantially constant torque is then obtained when the outer end of the spring is extended. It should be understood that the rotation axis about which the constant force spring **94** rotates is chosen so as to be orthogonal to the longitudinal axis of the rail **72** so that when the constant force spring **94** is unrolled, the outer end of the constant force spring **94** moves along a linear axis that is parallel to the rail **72**. As illustrated in FIG. 5, the outer end of the constant force spring **94** is provided with an aperture **114** for securing the outer end of the constant force spring **94** to an adapter **120**. In one embodiment, the adapter **120** is removably secured to the outer end of the constant force spring **94**.

(47) The adapter **120** connects to the outer end of the constant force spring **96** through a hole in the constant force spring **96**, and is secured with a screw. The securing means **84** is a rotating securing device, such as a quick connect adapter, which aligns a spring-loaded pin with the hole in the adapter **120**. When the pin and hole are aligned, the spring is released, thereby locking the securing means **84** to the adapter **120**. The user makes a rotating movement of the securing means **84** to align the pin with the hole. The securing means **84** is joined to the plate **82** with screws.

(48) While in the illustrated embodiment, the constant force spring **96** is rotatably secured to the casing **92** and the cover plate **94** via the shaft **102**, the spacer **104** and the bearing **106**, other embodiments may be possible as long as the constant force spring **96** is rotatably mounted relative to the elongated guiding structure **60**.

(49) Referring to FIG. 6, there is illustrated the two constant force springs **96** when secured to the connection structure **62**. Each adapter **120** to which the outer end of a respective constant force spring **96** is secured is removably secured to the connection plate **82** using the securing means **84**.

(50) As a result of the securing of the outer ends of the constant force springs **96** to the connection structure **62**, a displacement of the connection structure **62** along the rail **72** away from the spring assemblies **64** exerts a force on the constant force springs which unroll or uncoil. The uncoiling of each constant force spring **96** generates a restoring force on the connection plate **82** in a direction opposite to the displacement of the connection structure **62**, i.e., the restoring force is directed towards the spring assemblies **64**, as illustrated in FIGS. 7 and 7b. It should be understood that in FIGS. 7a and 7b, the connection structure **62** and the rail **72** are omitted to better see the constant force springs **96**. In FIG. 7a, the two constant force springs **96** are in a retracted position. In the retracted position, no downward force is exerted on the connection structure **62** and the distance between the adapters **120** and the casing **92** of its respective spring assembly **94** is minimal. FIG. 7b illustrates the constant force springs **96** when a downward force is exerted on the connection structure **62**. As a result of the downward force exerted on the connection structure **62**, the

connection structure **62** translates along the rail **72** away from the spring assemblies **64**. Since the adapters **120** are secured to the connection structure **62**, the downward force is transmitted to the adapters **120** and the adapters **120** also translate away from the spring assemblies **64**. Since the outer end of each constant force spring **96** is secured to its respective adapter **120**, the translation of the adapters **120** triggers a translation of the outer end of the constant force springs **96** along the elongated guiding structure **60** and therefore an uncoiling of the constant force springs **96**. In return, the uncoiling of the constant force springs **96** exerts a restoring force having a direction opposite to that of the downward force on the connection structure **62**. The restoring force acts against the downward force and prevents or reduces the downward translation of the connection structure **62** along the rail **72** of the elongated guiding structure **60**.

(51) In one embodiment, the adapters **120** and the securing means **84** may be omitted. In this case, the outer end of the constant force springs **96** may be directly secured to the connection plate **82**. For example, screws may be used for securing the constant force springs **96** to the connection plate **82**. In another example, the constant force springs **96** may be welded to the connection plate **82**.

(52) As mentioned above, the connection structure **62** further comprises a ball joint assembly **88** for securing the harness **14** to the connection plate **82** of the connection structure **62**. FIGS. **8** and **9** illustrate one embodiment of a ball joint assembly **88**. As illustrated in FIG. **8**, the ball joint assembly **88** allows for a rotation of the harness **14** about three different rotation axes **130**, **132** and **134** relative to the connection plate **82**. The first rotation axis **130** is orthogonal to the front face of the connection plate **82** of the connection structure **62**. The second and third rotation axes **132** and **134** are orthogonal together and parallel to the front face of the connection plate **82**.

(53) In the illustrated embodiment, the ball joint assembly **88** comprises a securing plate **140**, a central arm **142** and four lateral arms **144**. The securing plate **140** is designed to be secured to the connection plate **82** using screws, for example. The central arm **142** is secured to the securing plate **140** substantially at the center thereof. The lateral arms **144** are also secured to the securing plate **140**. In the illustrated embodiment, the lateral arms **144** are positioned at different angular positions about the central arm **142**. In one embodiment, the angular distance between two adjacent lateral arms **144** is constant so that the lateral arms are angularly equidistant.

(54) As illustrated in FIG. **9**, the central arm **142** comprises an elongated body **150** which extends between a first end secured to the securing plate **140** and a second end. A ball joint **151** is rotatably mounted at the second end of the elongated body **150**. The length of the central arm **142** is not variable. The ball joint **151** is further fixedly secured to the harness **14**, thereby allowing a rotation of the harness **14** about the three rotation angles **130**, **132** and **134** relative to the connection structure **62** as illustrated in FIG. **8**. In one embodiment, the length of the central arm **142** is fixed.

(55) Each lateral arm **144** comprises an elongated body **152** provided with a cavity **160**, a spring **161** a rod **154** partially inserted into the aperture **160**, a first pivotal connector **156** and a second pivotal connector **158**. The elongated body **152** extends longitudinally between a first end pivotally connected to the first pivotal connector **156** and a second end. The cavity **160** extends longitudinally from the second end of the elongated body **152** towards the first end along a given section of the length of the elongated body **152**. The spring **161** is inserted into the cavity **160**.

(56) The rod **154** extends longitudinally between a first end inserted into the cavity **160** of the elongated body **152** and abutting the spring **161**, and a second end pivotally connected to the second pivotal connector **158**. The first end of the rod **154** may translate within the cavity **160** of the elongated body **152**. The translation of the rod **154** into the cavity compresses the spring **161**. When the spring **161** achieves its maximal compression, the rod **154** can no longer translate into the cavity **160**.

(57) The first pivotal connector **156** is fixedly secured to the securing plate **140** such as using screws and the second pivotal connector **158** is fixedly secured to the harness **14**. The first pivotal connector **156** allows for a rotation of the elongated body **152** relative to the first pivotal connector **156** about a first rotation axis and the second pivotal connector **158** allows for a rotation of the rod

154 relative to the second connector **158** about a second rotation axis that is parallel to the first rotation axis of the first pivotal connector **156**.

(58) The four lateral arms **144** allows for limiting the rotation of the harness **14** relative to the securing plate **140** while maintaining the harness **14** in a rest position when not in use. When the harness **14** is rotated relative to the securing plate **140**, the rod **154** of at least one lateral arm **144** translates within the cavity **160** of its respective elongated body **152** towards its respective first pivotal connector **156** and/or the rod **154** of at least another lateral arm **144** translates within the cavity **160** of its respective elongated body **152** away from its respective first pivotal connector **156**. Within each cavity **160** of its respective elongated body **152**, a spring **161** provides a force opposed to the movement of the corresponding lateral arm **144**. Therefore, a force is applied against the rotation of the harness **14** relative to the securing plate **140**. This force increases as the angle of rotation from the original position of the harness **14** relative to the securing plate **140** increases. While the harness **14** is being rotated relative to the securing plate **140** and when a spring **161** reaches its maximal compression, the rotation of the harness is stopped since a rod **154** can no longer translate within the cavity of its elongated body **152**. Therefore, the four lateral arms **144** act as a limiting device for limiting the rotation of the harness **14** relative to the securing plate **140**, and therefore to the elongated guiding structure **60**, about any rotation axis **130**, **132**, **134**. Furthermore, when no force is exerted on the harness **14**, the springs **161** bias the rods **154** into a rest position relative to the cavities **160**, and therefore bias the harness **14** into a rest position. In one embodiment, the rest position for the harness **14** is chosen so as to be a straight position to facilitate the installation of the harness **14** on a user.

(59) Each pivotal connector **156** and pivotal connector **158** is positioned at an equal distance from the axis of the elongated body **150**. This distance is calculated to permit a maximum rotation between the harness **14** and the securing plate **140**, considering the length of the cavity of the elongated body **152**. The pivotal connector **156** and pivotal connector **158** are oriented such that each lateral arm **144** rotates directly towards or away from the axis of elongated body **150**.

(60) While the illustrated embodiment, the ball joint assembly **88** comprises four lateral arms **144** for limiting the rotation of the harness relative to the elongated guiding structure **60**, it should be understood that the number of lateral arms **144** may vary as long as the ball joint assembly **88** comprises at least three lateral arms **144**. Similarly, it should be understood that the position of the lateral arms **144** relative to the central arm **142** may vary and/or the orientation of the rotation axis of the first pivotal connector **156** relative to the position of the central arm **142** may vary.

(61) It should be understood that the lateral arms **144** are exemplary only and may be replaced with any adequate device or structure that limits the amplitude of the rotation of the harness relative to the plate **140** about at least one rotational axis.

(62) It should also be understood that the lateral arms **144** may be omitted.

(63) While in the illustrated embodiment it is mounted so that the ball joint **151** be mounted on the harness **14**, the ball joint assembly **88** may be reversely mounted, i.e., the central arm **142** may be mounted between the plate **140** and the harness **14** so that the ball joint be rotatably secured to the plate **140** and the elongated body **150** be secured to the harness **14**.

(64) As illustrated in FIG. **10**, a first connection head **164** is mounted to the proximal end of the elongated body **152** and a second connection head **166** is mounted at the distal end of the rod **154**. The first and second connection heads **164** and **166** are each provided with an aperture in which a pin is to be inserted for rotatably securing the first connection head **164** to a first pivotal connector **156** and the second connection head a respective second pivotal connector **158**, respectively.

(65) In one embodiment, the harness **14** comprises a rigid structure, such as metal structure, and foam pads, such as memory foam pads, are secured. The harness **14** is designed for gripping a user at the pelvis. Two straps covered with memory foam pads are attached to the metal structure and hang beneath the user so as to provide pelvic support. The straps are linked both to the front of the harness **14** and to the rear of the harness **14**. At least two additional straps covered with foam pads

may be passed in front of the user to provide trunk support. The straps are linked from the right side of the harness **14** to the left side of the harness **14**.

(66) While the illustrated ball joint assembly **88** offers three rotational degrees of freedom between the elongated guiding body **60** and the harness **14**, it should be understood that the ball joint assembly **88** may be modified to offer only one or two rotational degrees of freedom between the elongated guiding body **60** and the harness **14**.

(67) It should also be understood that the ball joint assembly **88** is exemplary only and that the illustrated ball joint assembly **88** may be replaced with an adequate assembly that provides at least one rotational degree of freedom between the elongated guiding body **60** and the harness **14** illustrated in FIG. **8**.

(68) It should further be understood that the ball joint assembly **88** may be replaced by any adequate device for fixedly securing the harness **14** to the trolley **80** of the connection structure **62** so that no degree of freedom exists between the harness **14** and the trolley **80**.

(69) In one embodiment, a constant force spring may be replaced by another one to adjust the restoring force applied to the connection structure **62**. In one embodiment, only a spring is replaced. In another embodiment, the spring assembly **64** comprising the shaft **102**, the spacer **104**, the bearing **106**, the spring **96** and the casing **92** is removed from the base plate **90** and another spring assembly **64** of which the spring **96** is configured for providing a greater or a smaller restoring force. The other spring assembly **64** is then mounted on the base plate **90** and connected to the adapter **120**.

(70) In one embodiment, the adapter **120** is fixedly secured to the outer end of the spring **96**. In another embodiment, the adapter **120** is removably securable to the outer end of the spring **96** so that the same adapter **120** can be used for connecting different springs **96** to the connection structure **62**. To secure the adapter **120** to the spring **96**, the outer end of the spring **96** is inserted into the adapter until the hole provided in the outer end of the spring **96** faces the hole provided in the adapter **120** and a fastener such as a screw is inserted into the two facing holes.

(71) In one embodiment, the securing means **84** is a rotating securing device, such as a quick connect adapter, which aligns a spring-loaded pin with the hole in the adapter **120**. When the pin and hole are aligned, the spring is released, locking the securing means **84** to the adapter **120**. In order to align the pin and the hole together, a rotation of the securing means **84** needs to be performed.

(72) In one embodiment, the weight support device is further provided with at least one sensor. For example, the weight support device may comprise two sensors. The first sensor may be configured for measuring the distance between the bottom of the connection plate **82** and the tube **22** and may be mounted on the tube **22**. The distance between the bottom of the connection plate **82** and the tube **22** provides information about the height of the connection plate **82**, i.e., about the position of the connection plate **82** relative to the tube **22** along the rail **72**. From the height of the connection plate **82**, the height of the harness **14** and therefore the user's vertical position can be determined. From the user's vertical position, the user's gait and fatigue can be determined to determine when the user begins to feel muscle fatigue since at this point in time, the user descends in the support. The second sensor may be configured for measuring the distance travelled by the user while using the walker and/or the displacement speed and may be mounted to one of the wheels **30-36**, such as on wheel **30** or **32**.

(73) In the same embodiment, the weight support device is further provided with a processing unit in communication with the sensor(s). The processing unit is configured for receiving measurement data from the sensor(s) and transmitting the measurement data to a computer machine. In one embodiment, the transmission and/or reception of the data is performed wirelessly. In one embodiment, the processing unit is further configured for performing data processing. For example, the processing unit may be configured for averaging the received measurement data and/or calculating the median of the received measurement data. The processing data may also be

configured for correcting the received measurement data by applying mathematical operations, such as additions and/or multiplications, on the received measurement data to correct for biases in the sensor(s). The processing unit may also further be configured for calculated secondary data from the received measurement data. The secondary data may be obtained by applying simple or complex mathematical operations on received measurement data, such as by applying multiplication by constants, integration, derivation, linear and nonlinear functions and matrix algebra on the received measurement data.

(74) In the following, there is described a further embodiment of a walking aid provided with a weight support device. While for the walker **10** the spring assembly **64** is mounted at the top of the U-shaped bar **20** so that the springs are directly connected to the connection body **62**, in the above-described walking aid the springs are indirectly connected to the connection body.

(75) FIGS. **11-14** illustrate one embodiment of a walking aid in the shape of a walker **200** provided with a frame **202**, a weight support device **204** and a harness **206**. It should be understood that the illustrated walker **200** is exemplary only and the weight support device **204** may be used in connection with any other adequate walker, walking aid, treadmill, or the like. As described in greater detail below, the weight support device **204** is mounted to the frame **202** and the harness **206** is mounted to the weight support device **204**.

(76) The frame **202** comprises a U-shaped bar or rod **210**, a transverse bar **212**, two bars **214** and **216**, four wheels **220**, **222**, **224** and **226**, two reinforcement arms **228** and **230**, and two support arms **232** and **234**. The wheels **220** and **222** are rotatably secured to a respective end **242**, **244** of the bar **214** while the wheels **224** and **226** are rotatably secured to a respective end **246**, **248** of the bar **216**.

(77) The U-shaped bar **210** extends between a first end **250** and a second end **252**. The first end **250** of the U-shaped bar **210** is mounted to the bar **216** adjacent the end **248** thereof and the second end **252** is mounted to the bar **214** adjacent the end **244** thereof.

(78) The U-shaped bar **210** comprises two straight and parallel portions **260** and **262** connected together by a curved portion **264**. The transverse bar **212** longitudinally extends between a first end that is mounted to the straight portion **260** of the U-shaped bar **210**, and a second end that is mounted to the straight portion **262** of the U-shaped bar **210**. The reinforcement arm **228** extends between a first end secured to the bar **214** at a location located between the two wheels **220** and **222**, and a second end secured to the straight portion **260** of the U-shaped bar **210**. The reinforcement arm **230** extends between a first end secured to the bar **216** at a location located between the two wheels **224** and **226**, and a second end secured to the straight portion **262** of the U-shaped bar **210**.

(79) The arm **232** is mounted to the straight portion **260** of the U-shaped bar **210** and projects towards the front side of the walker **200**, i.e., towards the wheels **220** and **224**. The arm **234** is mounted to the straight portion **262** of the U-shaped bar **210** and projects towards the front side of the walker **200**.

(80) The weight support device **204** is mounted to the curved portion **264** of the U-shaped bar **210** and the transverse bar **212**. It should be understood that any adequate method/mechanism for mounting the weight support device **204** to the frame may be used. For example, the weight support device **204** may be provided with an aperture for receiving the U-shaped bar **220** therein. In another example, the U-shaped bar **210** may comprise two portions, i.e., a first portion extending between the end **252** and another end secured to the weight support device **204**, and a second portion extending between the end **250** and another end secured to the weight support device **204**.

(81) In one embodiment, the weight support device **204** is mounted to the frame **102** so that when the walker **200** is deposited on a horizontal surface, the weight support device **204** extends vertically. In one embodiment, such a condition may be achieved when the longitudinal axis of the weight support device **204** is orthogonal to the plane passing by the rotation axes of the wheels **220-228**. In this case, the weight support device **204** is said to be mounted vertically so that the

force exerted by the weight support device **204** on the harness **206** is vertical, as explained below in greater detail.

(82) As described in greater detail below, the harness **206** is slidably mounted to the weight support device **204**.

(83) In operation, the harness **206** is secured to the back of a user and the weight support device **204** supports at least partially the weight of the user. As the user walks, the wheels **220-226** rotate and the walker **200** rolls and follows the user while continuously supporting at least partially the weight of the user.

(84) FIGS. **15-18** illustrate the weight support device **204** in a rest position. The weight support device **204** comprises an elongated body **300** extending along a longitudinal axis and being securable to the transverse bar **212** and the U-shaped bar **210** so that the top portion of the elongated body **300** extends above the U-shaped bar **210**.

(85) The elongated body **300** extends longitudinally between a first or bottom end **302** and a second or top end **304** and laterally between a front face **306** and a back face **308**. In the illustrated embodiment, the bottom end **302** of the elongated body **300** is mountable to the transverse bar **212** and the top end **304** extends over the U-shaped bar **210** when the elongated body **300** is mounted to the U-shaped bar **210**.

(86) The weight support device **204** further comprises a front trolley **320**, a spring assembly **322** and a connection assembly **326** mechanically connecting the front trolley **320** and the spring assembly **322** together. The front trolley **320** is movably mounted to the front face **306** of the elongated body **300** so as to move along the longitudinal axis of the elongated body **300**. In the illustrated embodiment, a front rail **330** is mounted on the front face **306** of the elongated body **300** and extends along at least a longitudinal section of the elongated body **300** so that the front trolley **320** may slide along the front rail **330**.

(87) As illustrated in FIGS. **16** and **18** the spring assembly **322** is mounted to the elongated body **300** on the back face **308** thereof. In the illustrated embodiment, the spring assembly **322** is mounted to the elongated body **300** adjacent the bottom end **302** thereof. However, the person skilled in the art will understand that the spring assembly **322** may be mounted at another location along the length of the elongated body **300**.

(88) The connection assembly **326** extends between a first end connected to the front trolley **320** and a second end connected to at least one spring contained into the spring assembly **322**. In operation and as described in greater detail below, a translation of the front trolley **320** towards the bottom end **302** of the elongated body **300** triggers an extension of the spring(s) contained in the spring assembly **322**. As a result of the extension of the spring(s), a restoring force is applied on the front trolley **320** in a direction opposite to the translation of the front trolley **320**, i.e., the restoring force is vertical towards the top end **304** of the elongated body **300**. In one embodiment, the connection assembly **326** is provided with a non-extendable length.

(89) A first section of the connection assembly **326** that is secured to the front trolley **320** faces the front face **306** of the elongated body **300**. A second section of the connection assembly **326** passes over the top end **304** of the elongated body **300**. A third section of the connection assembly **326** is connected to the spring assembly **322** and faces the front face **308** of the elongated body **300**.

(90) In the illustrated embodiment, the weight support device **204** comprises a pulley device **340** mounted at the top end **304** of the elongated body **300** and the connection assembly **326** engages the pulley device **340** so as to facilitate the motion of the connection assembly **326** over the top end **304** of the elongated body **300**. However, it should be understood that the pulley device **340** may be omitted. In this case, the top end **304** of the elongated body **300** may be provided with a curved shape so as to facilitate the motion of the connection assembly **326** thereover.

(91) It should be understood that at least the portion of the connection assembly **326** that passes over the top end **304** of the elongated body **300** during the translation of the front trolley **320** is flexible. In one embodiment, this portion of the connection assembly **326** is flexible and

substantially non-extendable.

(92) In the illustrated embodiment, the connection assembly **326** comprises a first flexible body **350** such as a first belt, strap, or the like, a back trolley **352**, a second flexible body **354** such as a second belt, strap, or the like, a first connector **356**, a second connector **358**, a third flexible body **360** such as a third belt, strap, or the like, and a third connector **362**.

(93) The first flexible body **350** is made of a flexible material and extends between a first end connected to the front trolley **320** and a second end mounted to the back trolley **352**. A first section of the first flexible body **350** extends from the front trolley **320** to the pulley device **340** facing the front face **306** of the elongated body **300**, a second section of the first flexible body **350** engages the pulley device **340**, and a third section of the first flexible body **350** extends from the pulley device **340** to the back trolley **352** facing the back face **308** of the elongated body **300**.

(94) The back trolley **352** is slidably mounted to the back face **308** of the elongated body and may translate along the longitudinal axis of the elongated body along at least a section thereof. In the illustrated embodiment, the weight support device **204** comprises a rail **364** mounted to the rear face **308** of the elongated body **300**. The rail **364** extends longitudinally along the longitudinal axis of the elongated body **300** and along a given section of the length thereof. The back trolley **352** is slidably mounted to rail **364** so as to translate therealong.

(95) The second flexible body **354** is made of a flexible material and extends between a first end connected to the back trolley **352** and a second end secured to the first connector **356**. The first and second connectors **356** and **358** are mating connectors, i.e., they are removably securable together.

(96) The third flexible body **362** is made of flexible material and extends between a first end secured to the second connector **358** and a second end secured to the third connector **362** which is secured to the springs as described below.

(97) Referring now to FIG. **20**, there is illustrated one embodiment of a spring assembly **322**. The spring assembly **322** comprising a casing **370** provided with an opening **372** on atop face thereof. Two constant force springs **374** and **376** are rotatable mounted to the casing **370** within the casing **370** and one end of the springs **376** and **387** is secured to the connector **362**.

(98) In the illustrated embodiment, the casing **370** is removably securable to the elongated body **330**. In this case, the weight support device **204** further comprises two curved plates or brackets **380** and **382** which are secured to the back face **308** of the elongated body and spaced apart so to form a slot therebetween. The casing **370** is further provided with a tongue **378** projecting from a front face thereof and the casing **370** is removable secured to the elongated body **300** by inserting the tongue **378** into the slot defined between the brackets **380** and **382**.

(99) In the illustrated embodiment, the spring assembly **322** further comprises spools **383** and **384** on which the constant force springs **374** and **376** are mounted, respectively, and the constant force springs **374** and **376** are fixedly held in place on the spools **383** and **384** thanks to spool caps **386** and **388**, respectively. The assemblies formed of the spools **383** and **384**, the springs **374** and **376** and the spool caps **386** and **388** are rotatably mounted into the casing **370** via bearings **390**. One end of the springs **374** and **376** is fixedly secured to its respective spool **383**, **384** and the end of the springs **374** and **376** that projects therefrom is secured to the connector **362**. As a result, when a force directed away from the spring assembly **322** is exerted onto the connector **362**, the spools **383** and **384** rotate and the springs **374** and **376** are unwound.

(100) Referring back to FIGS. **15-19**, when a force directed towards the end **302** of the elongated body **300** is applied on the front trolley **320**, the front trolley moves downwards, i.e. towards the end **302** of the elongated body **300**. Since the back trolley **352** is connected to the front trolley **320** via the first flexible body **350** and the first flexible body **350** is longitudinally non-extendable, i.e., its length is substantially constant, a downwards translation of the front trolley **320**, i.e., a translation of the front trolley towards the end **302** of the elongated body **300**, triggers a translation of the back trolley **352** in the opposite direction, i.e., towards the end **304** of the elongated body **300**. Since the end of the constant force springs **374** and **376** is secured to the third connector **362**,

the third connector **362** is connected to the back trolley **352** via the second and third flexible bodies **354** and **360** and the connectors **356** and **358**, and the second and third flexible bodies **354** and **360** have substantially constant length, a translation of the back trolley towards the end **304** of the elongated body **300** triggers an unwinding of the springs **374** and **376** which extend partially outside of the casing **70**. As a result, the springs **374** and **376** exert a restoring force on the front trolley **320** in a direction opposite to the translation of the front trolley **320**, i.e., towards the end **302** of the elongated body **300**. The restoring force allows for supporting at least partially the weight of the user.

(101) While FIGS. **15-18** illustrate the weight support device **204** when the front trolley **320**, the back trolley **352** and the springs **374** and **376** are each in a rest position, i.e., when no downwards force is applied to the front trolley **320** and the springs are wound, FIGS. **21-24** illustrate the weight support device **204** when the front trolley **320** is in an active position, i.e. when a downwards force is exerted on the front trolley **320**. When the front trolley **320** is in an active position, the position of the front trolley **320** is located between its rest position and the bottom end of the front rail **330**, i.e., the end of the front rail **330** facing the end **302** of the elongated body **300**. Similarly, the back trolley **352** is in an active position which is located between its rest position and the top end of the back rail **364**, i.e., the end of the back rail **364** facing the end **304** of the elongated position. When the front trolley **320** is in an active position, the springs **374** and **376** are partially unwound and partially extend from the casing **370**.

(102) In the illustrated embodiment, the front trolley comprises a connection plate **392** for attachment of the harness **206** thereto. However, it should be understood that any adequate mechanism/method may be used for mounting the harness **206** to the front trolley **320**.

(103) FIGS. **25** and **26** illustrate one embodiment of the harness **206** designed and shaped to grip the user's pelvis. The harness **206** comprises a mounting portion **400** and a cushion portion **402** which are pivotally connected together by a pivot assembly **406**.

(104) The mounting portion **400** comprises a base plate **410** securable to the connection plate **392** via four standoffs **412** secured to the back of the base plate **410**. The mounting portion **400** further comprises a roller bearing **414** and a connection plate **416**. The roller bearing **414** is mounted between the front face of the base plate **410** and the back face of the connection plate **416** so that the connection plate may rotate relative to the base plate **410** about a rotation axis orthogonal to the base plate **410**.

(105) The cushion portion **402** comprises a central plate **420**, two curved arms **422** and **424**, a central cushion **426** and two foam covered plates **428** and **430**. The central cushion **426** is mounted to the front face of the central plate **420**. The arms **422** and **424** are slidably mounted to the back face of the central plate **420** each via a sliding mechanism and they project laterally from a respective lateral side from the central plate **420**. As a result, the length of the portion of each arm **422**, **424** that projects from the central plate **420** is adjustable. The foam covered plate **428** is secured at the end of the arm **422** that is opposite to the central plate **420** and the foam covered plate **430** is secured at the end of the arm **424** that is opposite to the central plate **420**. The cushion **426** is mounted on the front face of the central plate **420**.

(106) The pivot assembly **406** is secured to the front face of the connection plate **416** and the back face of the central plate **420**. The pivot assembly comprises an upper pivot module **440**, a middle pivot module **442**, a lower pivot module **444**, two bearings **446** and **448** and a shoulder screw **450**. Each of the upper, middle and lower pivot modules **440**, **442** and **444** is provided with an aperture extending therethrough for receiving the shoulder screw **450** therein. The upper and lower pivot modules **440** and **446** are secured to the front face of the connection plate **416** while the middle pivot module **442** is secured to the back face of the central plate **420** between the upper and lower pivot modules **440**. The bearing **446** is mounted between the upper and middle pivot modules **440** and **442** while the bearing **448** is mounted between the middle and lower pivot modules **442** and **444**. The pivot modules **440**, **442** and **444** and the bearings **446** and **448** are aligned so that their

respective apertures are aligned, thereby allowing the insertion of the shoulder screw **450** into the apertures of the different components.

(107) The pivot assembly **406** allows for a rotation of the cushion portion **402** relative to the mounting portion **400** about a rotation axis that is orthogonal to the rotation axis of the roller bearing **414**. As a result the cushion portion of the harness **406** may rotate about two rotation axes relative to the front trolley **320**, i.e., about a first rotation axis orthogonal to the front face of the connection plate **392** of the front trolley **320** and a second rotation axis orthogonal to the first rotation axis.

(108) In operation, the harness **206** is mounted around the pelvis of the user and the arms **422** and **424** are adjusted so that the harness **206** firmly engages the pelvis of the user. The user may then walk and the walker rolls and follows the user. Furthermore, the weight support device **204** of the walker **200** allows for supporting at least partially the weight of the user as the user walks. When due to the weight of the user, the harness **206** moves downwards, thereby triggering a translation of the front trolley towards the end **302** of the elongated body **330** and the unwinding of the constant force springs **374** and **376**, an upward restoring force is exerted on the harness **206** by the constant force springs **374** and **376**, thereby allowing a partial support of the weight of the user.

(109) It should be understood that the weight support device **204** may be modified. For example, while it is removably securable to the elongated body **300**, the spring assembly **322** may be permanently secured to the elongated body. For example, the casing **370** of the spring assembly **322** may be welded to the elongated body. In this case, the connection assembly may comprise a single flexible body extending between a first end secured to the trolley **320** and a second end secured to springs **374** and **376**.

(110) In another example, the connector **352** and the flexible body **354** may be omitted. In this case, the flexible body **350** is directly secured to the connector **356**.

(111) It should also be understood that the number of constant force springs may vary as long as the weight support device **204** comprises at least one constant force spring.

(112) In an embodiment in which the spring assembly **322** is removably securable to the elongated body **300**, the spring assembly **322**, the connector **362**, the flexible body **360** and the connector **358** form an interchangeable spring unit. In this case, different interchangeable spring units comprising different springs, i.e., springs capable of generating different restoring forces, can be removably connected to the elongated body **300**. For a given user having a given weight, an adequate interchangeable spring unit is chosen based on the restoring force that can be generated by its springs, i.e., the restoring force must be sufficient to at least partially support the weight of the given user. The casing **370** of the adequate interchangeable spring unit is then secured to the elongated body **300** and the connector **358** of the adequate interchangeable spring unit is connected to the connector **356**, and the walker **200** may then be used by the given user. If the walker **200** is to be used by another user, another interchangeable spring unit is chosen based on its springs and the weight of the other user, and removably secured to the elongated body as described above.

(113) In one embodiment, the walker **200** is further provided with a data acquisition system for retrieving real-time objective data on the user's rehabilitation by measuring the displacement of the walker **200**. The data may include the position of the walker **200** and their speed of movement, as well as the height and orientation of the user's hips. Subsequently, this data can be accessed by the user, a health professional, and/or the like to assess the user's progress and guide the user's training and rehabilitation program.

(114) In one embodiment, the data acquisition system comprises three main subsystems, i.e., sensors, a processing unit and a data transmission unit.

(115) The sensors first make it possible to collect the data by targeting different characteristics of the use of the walker.

(116) In one embodiment, the sensors comprise an orientation sensor, a height sensor, a displacement sensor, and/or at least one pressure sensor.

(117) The orientation sensor is configured for measuring the movements of the user's pelvis. The orientation sensor may comprise a gyroscope secured to the central plate **420** of the walker **200**.

(118) The height sensor is configured for measuring the height of the harness and therefore the user's pelvis height. For example, the height sensor may be mounted at the lower end of the rail **330** and be configured to measure the distance between the lower end of the rail **330** and the plate **392** from which the height of the harness, and therefore the user's pelvis height, can be determined. The height of the user's pelvis can then be correlated to the muscular fatigue felt by the user since the latter will tend to sag under the effect of fatigue.

(119) The displacement sensor is configured for measuring the travelled distance, the speed and/or the acceleration of the walker **200**. For example, the displacement sensor may comprise a hall effect sensor paired with a magnet mounted to one of the wheels **220-226** so as to measure the number of revolutions of the wheel. This data can be used to calculate the displacement, speed and acceleration of the walker **200**, and therefore of the user, in real time.

(120) The pressure sensor(s) is(are) configured to measure the lateral forces applied by the user on the harness and may be inserted into at least one of the plates **428** and **430** and/or into the cushion **426** or behind the cushion **426**. As with the data from the orientation sensor, these efforts can make it possible to deduce the movements of the user's pelvis for the purpose of analyzing the gait pattern.

(121) It should be understood that other types of sensors may be integrated into the walker **200**.

(122) The processing unit is configured for gathering data from the sensors and stores the data into a memory. The processing unit may be configured to perform different types of operations such as cleaning and correcting data, transforming data into different units, applying algorithms to calculate secondary data, processing data for better presentation to system users, etc. The processing unit may comprise a microcontroller, a microprocessor, a central processing unit, a graphical processing unit, and/or the like.

(123) In one embodiment, the use of at least one constant force spring **96, 374, 376** allows for applying a constant force on the harness **14, 206** independently of the deformation of the constant force spring **96, 374, 376**. This allows for offering a constant support to the user over the amplitude of displacement of the trolley **80, 320**.

(124) In one embodiment, the wright support device **12, 204** is mounted on the walker **10, 200** so that the elongated body **70, 300** extends vertically when the walker **10, 200** is deposited on a horizontal surface. Such a configuration allows for a vertical translation of the trolley **80, 320**, which in turns apply a vertical force on the user that is opposite to gravity. Such a configuration allows for decreasing or substantially eliminating the force components that could induce a loss of balance for the user, thereby increasing the stability of the user. Furthermore, such a configuration allows a substantially constant support for users having different heights.

(125) For example, for the walker **200** in which the wheels **220-226** are identical, the vertical attachment of the elongated body **300** is achieved by having the longitudinal axis of the elongated body **300** being orthogonal to the plane passing by the rotation axes of the wheels **220-226**.

(126) In one embodiment, the positioning of the spring assembly **322** at the bottom of the elongated bottom allows for lowering the center of gravity of the walker **200**, thereby increasing the stability of the walker.

(127) In one embodiment, the positioning of the spring assembly **322** on the back face of the elongated body **300** allows for moving the spring assembly **322** away from the user, thereby reducing risks of injury for users when the springs **374** and **376** moves out or into the casing **370**.

(128) In one embodiment of the spring assembly comprising two constant force springs such as the spring assembly **322**, the constant force springs are positioned so that their section that may exit the casing be positioned back to back, as illustrated in FIGS. **22** and **24**. Such a configuration allows for improving the performance of the springs. The back to back positioning of the springs allows for decreasing their deformation as they extend from the casing, thereby improving their lifespan,

in addition to reducing the space required for accommodating the springs within the casing.
(129) Modifications and improvements to the above-described implementations of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting.

Claims

1. A weight support device for use with a walking aid to at least partially support a weight of a user, the weight support device comprising: an elongated guiding body extending between a first and a second end along a longitudinal axis and laterally between a front face and a rear face, the elongated guiding body being mountable to the walking aid, the longitudinal axis extending substantially vertically when the elongated guiding body is mounted to the walking aid and the walking aid is positioned on a horizontal surface and the first end being higher than the second end when the elongated guiding body is mounted to the walking aid; a connection body slidably mounted to the elongated guiding body for translating along the longitudinal axis of the elongated guiding body; at least one constant force spring rotatably mounted to the elongated guiding body, an outer end of the at least one constant force spring being connected to the connection body; and a body securing device for engagement with the user, the body securing device being mounted to the connection body, wherein when the elongated guiding body is mounted to the walking aid, the at least one constant force spring exerts a restoring constant force on the connection body when the connection body moves away from the first end of the elongated guiding body so as to compensate at least partially for the weight of the user.
2. The weight support device of claim 1, wherein the connection body is mounted on the front face of the elongated guiding body.
3. The weight support device of claim 2, wherein the at least one constant force spring is mounted at the first end of the elongated guiding body.
4. The weight support device of claim 3, wherein the outer end of the at least one constant force spring is secured to the connection body.
5. The weight support device of claim 4, wherein the outer end of the at least one constant force spring is removably secured to the connection body.
6. The weight support device of claim 3, wherein the elongated guiding body comprises a rail extending along the longitudinal axis and the connection body comprises a trolley slidably mounted to the rail.
7. The weight support device of claim 6, wherein the connection body further comprises a connection plate mounted to the trolley, the outer end of the at least one constant force spring being connected to the connection plate.
8. The weight support device of claim 1, wherein the connection body comprises a joint assembly securable to the body securing device.
9. The weight support device of claim 2, wherein the at least one constant force spring is mounted on the rear face of the elongated guiding body, the weight support device further comprising a connection assembly extending between a first assembly end connected to the connection body and a second assembly end connected to the outer end of the at least one constant force spring, the connection assembly comprising at least a flexible section to engage the first end of the elongated guiding body.
10. The weight support device of claim 9, wherein the elongated guiding body comprises a front rail extending along the longitudinal axis and the connection body comprises a front trolley slidably mounted to the front rail.
11. The weight support device of claim 10, wherein the connection assembly comprises an elongated flexible body connected between the outer end of the at least one constant force spring and the connection body.

12. The weight support device of claim 10, wherein the connection assembly comprises a first connector and a second connector removably securable together, the first connector being connected to the connection body and the second connector being connected to the outer end of the at least one constant force spring.
 13. The weight support device of claim 12, wherein the elongated guiding body comprises a rear rail extending along the longitudinal axis and the connection assembly further comprises a rear trolley slidably mounted to the rear rail, the rear trolley being connected to the connection body and the first connector.
 14. The weight support device of claim 10, wherein the front trolley comprises a connection plate.
 15. The weight support device of claim 14, wherein the connection body comprises a first rotational connection mounted to the connection plate and a second rotational connection mounted to the first rotational connection, the body securing device being mounted to the second rotational connection.
 16. The weight support device of claim 9, wherein the at least one constant force spring is mounted adjacent the second end of the elongated guiding body.
 17. The weight support device of claim 1, wherein the body securing device comprises a harness.
 18. A walking aid system, comprising: a walking aid; and a weight support device of claim 1, the weight support device being secured to the walking aid so that the longitudinal axis extends substantially vertically when the walking aid is positioned on the horizontal surface.
 19. The walking aid system of claim 18, wherein the walking aid comprises one of a walker and a treadmill.
-